

Program9: Organize Kubernetes Pods Using Labels and Namespaces

Project Structure

DEVOPSLAB

```
--program9
|--namespaces.yaml
|--pods.yaml
```

Step 1: Create a file for Namespaces as namespaces.yaml

Step 2: Create a file for Pods as pods.yaml

Step 3: Apply the namespaces and pod yaml file

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/program9
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~$ cd DEVOPSLAB
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB$ mkdir program9
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB$ cd program9
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program9$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🔧 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📡 Pulling base image v0.0.48 ...
🔄 Updating the running docker "minikube" container ...
❗ Failing to connect to https://registry.k8s.io/ from inside the minikube container
💡 To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
📦 Using image gcr.io/k8s-minikube/storage-provisioner:v5
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   📦 Using image docker.io/kubernetesui/dashboard:v2.7.0
   📦 Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, dashboard, default-storageclass

❗ /usr/local/bin/kubectl is version 1.28.4, which may have incompatibilities with Kubernetes 1.34.0.
   📦 Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏁 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program9$
```

```
GNU nano 7.2 namespaces.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step4: Verify Pods by namespace

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/program9
GNU nano 7.2 pods.yaml *
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
```

Help Write Out Where Is Cut Execute Location M-U Undo M-A Set Mark
Exit Read File Replace Paste Justify Go To Line M-E Redo M-6 Copy

Step 5: Apply the namespaces and pod yaml files

Step 6: Verify Pods by namespace

Step 7: Check pod status

Step 8: Filter Pods by label

-List all frontend pods in dev namespace

-List all backend pods in prod namespace

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program9$ nano pods.yaml
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program9$ kubectl apply -f namespaces.yaml
namespace/dev1 created
namespace/prod1 created
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program9$ kubectl apply -f pods.yaml
pod/frontend-pod1 created
pod/backend-pod1 created
pod/frontend-pod1 created
pod/backend-pod1 created
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   5 (5m34s ago)  11h
frontend-pod  1/1     Running   5 (5m34s ago)  11h
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   5 (5m54s ago)  11h
frontend-pod  1/1     Running   5 (5m54s ago)  11h
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   5 (6m9s ago)  11h
frontend-pod  1/1     Running   5 (6m9s ago)  11h
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program9$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6cfcb5c844-b95pq  1/1     Running   0           3m19s
nginx-deployment-6cfcb5c844-c9d4n  1/1     Running   0           3m19s
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   5 (7m37s ago)  11h
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   5 (8m6s ago)  11h
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program9$
```